ALMA MATER STUDIORUM - UNIVERSITÀ DI BOLOGNA

At the Edge of Conservation: Livelihoods & Household Welfare in Maasai-Associated Areas of Northern Tanzania

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Motivation

- **Conservation vs. Pastoralism:** Protected areas restrict grazing mobility and land access. Costs may not appear first in consumption, but in herd management and livestock health
- **Two policy shocks:** 2009 Wildlife Conservation Act (legal consolidation) and 2013 anti-poaching enforcement (reported cattle seizures, house burnings)
- **A measurement problem:** Ethnicity is unobserved in modern Tanzanian surveys. This is a legacy of Nyerere-era nation-building. Maasai exposure must be reconstructed spatially

Data: **Tanzania National Panel Survey (NPS), Waves 1–4**



Empirical Strategy

Research question: Among households in the reserve risk set, did conservation exposure differentially affect those in Maasai-associated areas?

$$Y_{iwt} = \alpha + \beta_1(T_i \times Post2009_w) + \beta_2(T_i \times Post2013_w) + \delta T_i + X'_{iwt}\gamma + \lambda_w + \mu_d + \theta Distance_i + \rho Inside_i + \varepsilon_{iwt}.$$

β_1 = differential shift after 2009 **β_2** = additional shift after 2013 **Total post-2013** = $\beta_1 + \beta_2$

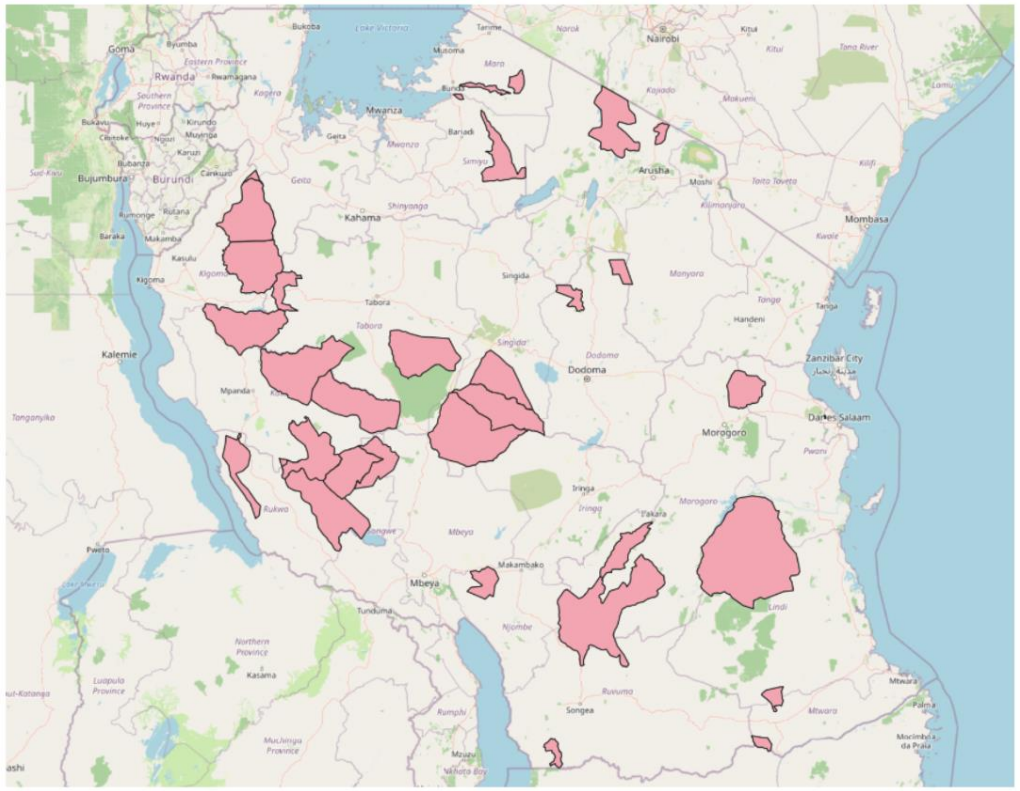
Key Definitions

Risk Set: within 75 km of a game-reserve boundary

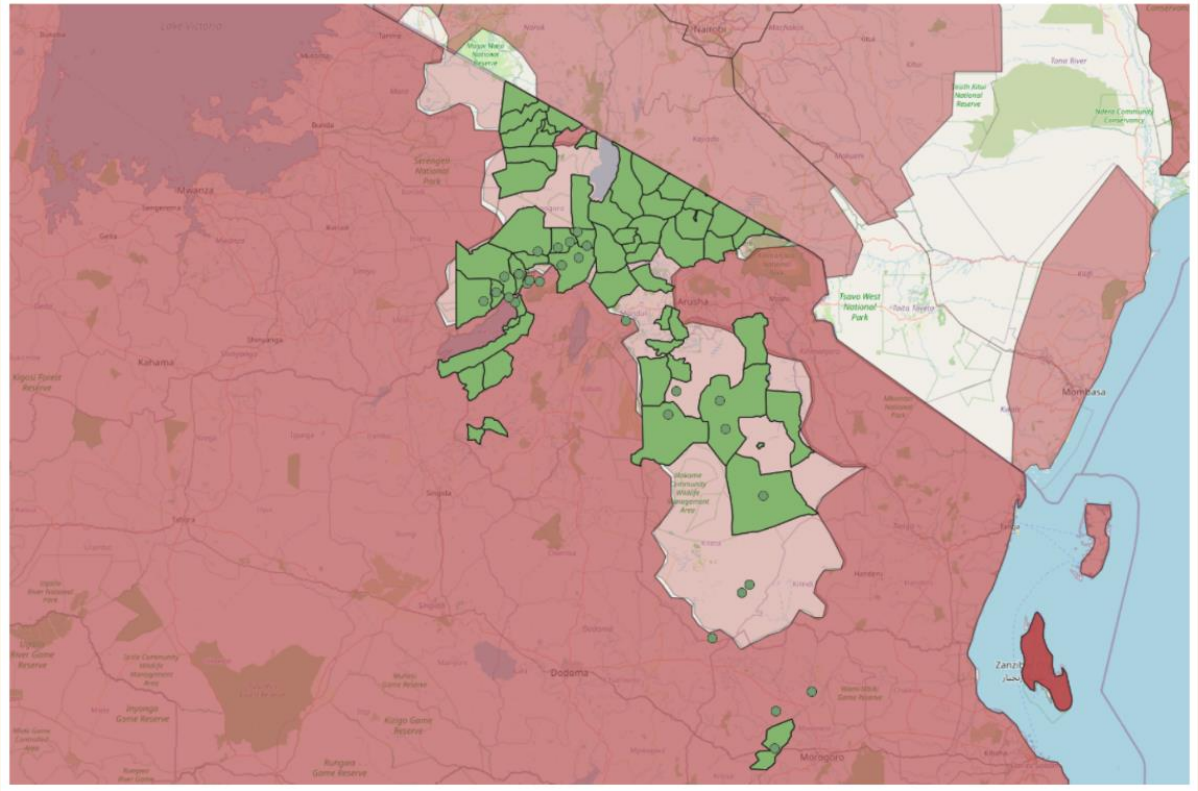
Treatment (T): within 25 km of Maasai-associated localities (validated areas)

Repeated cross-sections, wave & district fixed effects, household controls, EA-clustered standard errors.

Game Reserve Risk Set & Maasai-Association



Game Reserve Shapefile



Reconstructed Maasai exposure (green) vs. GeoEPR historical benchmark (pink)

Treatment layer built from documentary evidence and manual GIS validation: an area-based proxy for exposure, not individual ethnicity

Main Regression Results

Table 2: Main effects on welfare and herd dynamics

Outcome	Post-2009	Additional post-2013	Total post-2013	Obs.	EA clusters
Log consumption per capita	-0.083 (0.081) [0.300]	-0.025 (0.087) [0.785]	-0.108 (0.120)	8,850	257
Log consumption per adult equivalent	-0.095 (0.079) [0.245]	0.012 (0.085) [0.891]	-0.083 (0.115)	8,850	257
Poverty indicator	0.095** (0.045) [0.080]	-0.031 (0.045) [0.513]	0.065 (0.064)	8,959	261
Net cattle change	-2.045* (1.076) [0.046]	1.497 (1.067) [0.202]	-0.549 (0.444)	5,029	154
Any cattle decline	0.174*** (0.059) [0.007]	-0.068 (0.068) [0.331]	0.106*** (0.036)	5,029	154
Cattle exit	0.079** (0.031) [0.057]	-0.076** (0.031) [0.056]	0.003 (0.021)	1,124	77
Cattle loss share	0.078* (0.041) [0.094]	-0.047 (0.041) [0.303]	0.031 (0.031)	1,124	77
Total livestock per adult equivalent	1.292 (1.751) [0.603]	2.115* (1.213) [0.118]	3.406 (2.233)	6,380	160

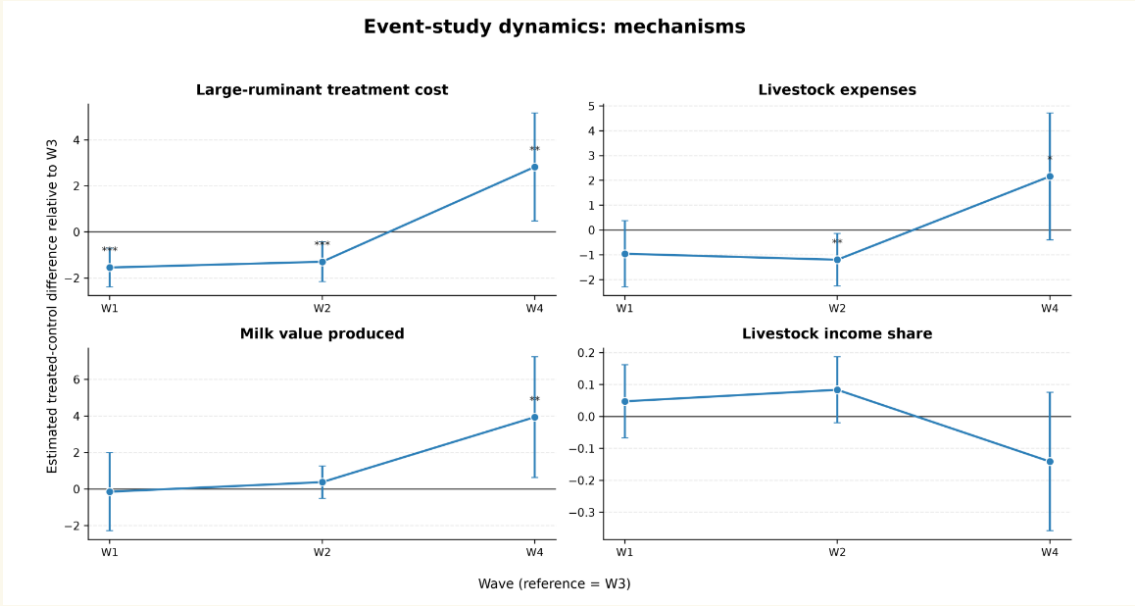
Takeaway: poverty rises after 2009; cattle dynamics deteriorate (decline, exit, loss share); consumption effects stay negative but statistically imprecise

Notes: Parentheses report the EA-clustered standard error. Square brackets report the wild-cluster-bootstrap p-value, where available. Significance stars are based on ordinary clustered p-values (not the bootstrap): * p<0.10, ** p<0.05, *** p<0.01

Mechanisms & Timing

Outcome	Post-2009	Add. post-2013	Total post-2013
Large-ruminant treatment cost	0.979***	3.391***	4.370***
Livestock expenses	0.431	2.682**	3.113**
Large-ruminant disease	-0.006	0.152**	0.147*
Milk value produced	0.310	3.775**	4.085***
Livestock income share	-0.009	-0.179*	-0.188

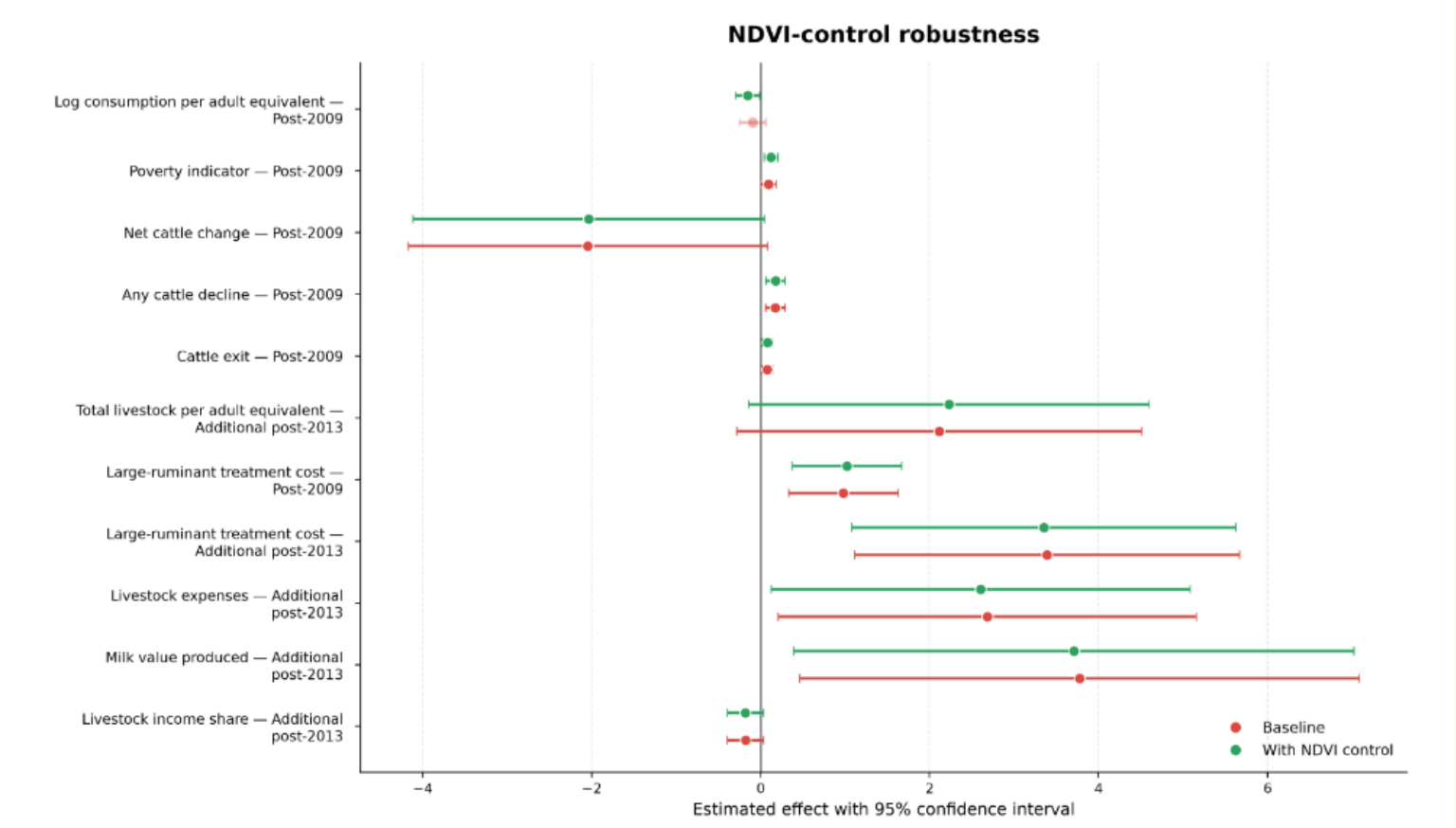
Not "herd collapse" but a costly transformation under pressure



Cost and productivity mechanisms rise sharply only after 2013 (Wave 4): households stay engaged in livestock, but at higher cost and fragility

Climate Control

Could this just be drought, not conservation? NDVI is added as a robustness control for local vegetation conditions (EA-wave level, 1 km scale)



Result: poverty and livestock-stress coefficients remain stable, the poverty effect even strengthens, weakening a purely climate-based explanation

Limitations

Non-random exposure	Maasai-associated areas may have unobserved time-varying differences vs. the rest of the risk set
Spatial proxy	Treatment is locality-based (area-level); it does not identify individual household ethnicity
Limited pre-period	Only one true pre-2009 wave: parallel trends cannot be fully tested for the 2009 shock
Unobserved migration	Repeated cross-sections may understate displacement if whole households left treated areas

Conclusion & Policy Implication

- **Livelihood stress:** Conservation costs concentrate in pastoral margins (animal health, input costs, herd dynamics) more than in aggregate consumption
- **Costly adaptation:** Households remain engaged in livestock but face higher fragility, not simple herd collapse
- **Policy implication:** Evaluation should go beyond consumption. Target veterinary access and animal-health support to build pastoral resilience near protected areas

Thank you for your attention - Questions?

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